

HTML & CSS Basics

- **Inline CSS:** style directly in tag → `<div style="font-family:Arial; font-size:16px; font-style:italic; font-weight:bold; text-decoration:underline; color:red; background-color:yellow;">Essay text</div>`
- **Internal CSS:** inside `<style>` in `<head>`. Example:

CSS

```
input { border:2px solid blue; margin:0 10px; }
```

```
img { border:3px solid red; border-radius:50%; margin-top:20px; }
```

- **iframe:** `<iframe src="page.html"></iframe>` → opens hyperlink inside iframe.

JavaScript Fundamentals

- **var vs let:**
 - `var` → function-scoped, hoisted.
 - `let` → block-scoped, safer for modern JS.
- **Functions:** normal, arrow, anonymous, callback, constructor, predefined (like `parseInt`, `Math.random`).
- **Varargs:** use `...args` to accept variable number of parameters.

JavaScript Assignments

- **Table of a number:**

js

```
function table(n){ for(let i=1;i<=10;i++){ console.log(n*i); } }
```

- **Pattern:**

Code

```
* * *
```

```
* *
```

```
*
```

→ nested loops decreasing stars.

- **Square of input:** <input type="number" required> + button → JS calculates $n*n$ and displays.
- **Array add/display:** push values into array, display with .
- **Email validation:** regex like `/^[^\s@]+@[^\s@]+\.[^\s@]+$/`.
- **Radio buttons:** uppercase/lowercase/as-is → use `.toUpperCase()`, `.toLowerCase()`.

DOM Manipulation

- Change paragraph color/font: use `document.getElementById("p1").style.color = "red";`.
- Currency converter: dropdowns for from/to, conversion logic ($₹80 = \$1$).
- Arrays.html: add, insert, remove elements in array → show in <p>.

Login & Forms

- **Login.html:** validate username/password, password length 8–12. If correct → redirect to Home.html.
- **NewUser.html:** form with multiple fields, languages checkboxes.
- **ChangePassword.html:** validate new password length.
- **AboutUs.html:** resume with table, list, description list.
- **index.html:** hyperlinks to all pages.

Node.js Concepts

- **REPL:** Read-Eval-Print-Loop → run JS directly in terminal.
- **Functions with callbacks:**

js

```
function result(a1,a2,callback){ console.log(callback(a1,a2)); }
```

- **Nested callbacks:** callback inside callback → “callback hell”.
- **Error handling in divide/multiply:** if divisor = 0 → return error.
- **Promises:** `new Promise((resolve,reject)=>{ ... })`.
- **Execution order:**
 - `console.log()` → runs first.

- Normal function → synchronous.
- setTimeout → goes to event queue.
- Promise → microtask queue → runs before timeout.
- **Async/Await:** waits for promise resolution.
- **Employee class:** constructor, getters/setters, display method.
- **File system (fs):**
 - readFileSync, readFile
 - writeFileSync, writeFile
 - appendFileSync
 - mkdirSync, renameSync, unlinkSync, existsSync.

JSON & Objects

- Convert object → JSON: JSON.stringify(obj).
- JSON → object: JSON.parse(jsonString).
- Object destructuring:

js

```
let {id,name,grade} = student;
```

- Prototype: add methods to objects via Object.prototype.

Arrays & Loops

- Array methods: filter, map, sort.
- Even numbers: arr.filter(x=>x%2===0).
- Odd numbers: arr.filter(x=>x%2!==0).
- For-in loop: iterate object properties.

jQuery

- Button click:

js

```
$("#btn").click(()=>$("#p1").text("Hello"));
```

- Factorial on change:

js

```
$("#num").change(()=>{ ... });
```

- Animation: `$("#div").animate({left:"200px",width:"100px",height:"100px"}).`
- Stop animation: `$("#div").stop().`

AJAX

- Load data:

js

```
$.get("https://fakestoreapi.com/products",(data)=>{
```

```
    // display in table
```

```
});
```

- POST method:

js

```
$.post("https://jsonplaceholder.typicode.com/todos",{title:"abc"},(res)=>{console.log(res)});
```

Extra Assignments

- **Password Generator:** random string using `Math.random()` and String methods.
- **Username Generator:** first 2 chars of firstname + age + last 2 chars of lastname.

Quick Exam Tips

- **HTML** → structure, tags, attributes.
- **CSS** → inline, internal, external.
- **JS** → variables, functions, DOM, events, arrays, objects, JSON.
- **Node.js** → callbacks, promises, async/await, fs module.
- **jQuery** → selectors, events, animation.
- **AJAX** → async data load, GET/POST.
- **Validation** → HTML5 attributes (required, minlength, maxlength), regex for email.

BRIEF:

HTML & CSS

Key Points

- Inline CSS: style directly in tag → `<div style="font-family:Arial; font-size:16px; font-style:italic; font-weight:bold; text-decoration:underline; color:red; background-color:yellow;">Essay</div>`
- Internal CSS: inside `<style>` in `<head>`.
- iframe: `<iframe src="page.html"></iframe>` opens hyperlink inside iframe.
- Forms: `<input type="text" required minlength="8" maxlength="12">`.

Cross Q&A

- Q: Difference between inline and internal CSS? A: Inline is applied directly in the tag, internal is inside `<style>` in the HTML file.
- Q: How to open a link inside the same page? A: Use `<iframe>` and set hyperlink target to the iframe.

JavaScript Fundamentals

Key Points

- `var` → function scoped, hoisted.
- `let` → block scoped, safer.
- Functions: normal, arrow, anonymous, callback, constructor.
- Varargs: `function sum(...args){ return args.reduce((a,b)=>a+b); }`.

Cross Q&A

- Q: Difference between `var` and `let`? A: `var` is function scoped and hoisted, `let` is block scoped and not hoisted.
- Q: What is a callback? A: A function passed as an argument to another function, executed later.

JavaScript Assignments

Key Points

- Table of number: loop from 1–10.
- Pattern: nested loops decreasing stars.
- Square: input type number, calculate $n*n$.
- Array add/display: push values, show in `<ul>`.
- Email validation: regex `/^[^\s@]+@[^\s@]+\.[^\s@]+$/`.
- Radio buttons: `.toUpperCase()`, `.toLowerCase()`.

Cross Q&A

- Q: How to validate email in JS? A: Use regex pattern.
- Q: How to display array in UL? A: Loop through array and append `<li>` elements.

DOM Manipulation

Key Points

- Change style: `element.style.color="red"; element.style.fontWeight="bold";`.
- Currency converter: dropdowns, conversion logic ($₹80 = \$1$).
- Arrays.html: add, insert, remove elements in array.

Cross Q&A

- Q: How to change paragraph color on button click? A: Use `document.getElementById("p1").style.color="blue";`.

Login & Forms

Key Points

- Login.html: validate username/password, password length 8–12.
- NewUser.html: form with multiple fields.
- ChangePassword.html: validate new password.
- AboutUs.html: resume with table, list, description list.
- index.html: hyperlinks to all pages.

Cross Q&A

- Q: How to validate password length in HTML? A: Use minlength and maxlength attributes.
- Q: How to redirect after login? A: Use JS window.location="Home.html";.

Node.js Concepts

Key Points

- REPL: interactive shell for Node.
- Callbacks: result(a1,a2,callback).
- Nested callbacks → callback hell.
- Error handling: check divisor = 0.
- Promises: new Promise((resolve,reject)=>{}).
- Async/Await: waits for promise resolution.
- Execution order: console.log → synchronous → Promise → setTimeout.
- Employee class: constructor, getters/setters, display.
- fs module: read/write/append files, mkdir, rename, unlink.

Cross Q&A

- Q: What is callback hell? A: Nested callbacks making code hard to read.
- Q: Difference between Promise and async/await? A: Promise handles async with then/catch, async/await makes code look synchronous.
- Q: How to read file asynchronously? A: fs.readFile("file.txt","utf8",(err,data)=>{});.

JSON & Objects

Key Points

- Convert object → JSON: JSON.stringify(obj).
- JSON → object: JSON.parse(jsonString).
- Destructuring: let {id,name} = student;.
- Prototype: add methods to object.

Cross Q&A

- Q: How to convert object to JSON? A: Use `JSON.stringify()`.
- Q: What is destructuring? A: Extracting properties into variables.

Arrays & Loops

Key Points

- Array methods: `filter`, `map`, `sort`.
- Even numbers: `arr.filter(x=>x%2===0)`.
- Odd numbers: `arr.filter(x=>x%2!==0)`.
- For-in loop: iterate object properties.

Cross Q&A

- Q: How to filter even numbers? A: Use `filter` with condition `x%2===0`.
- Q: Difference between for-in and for-of? A: for-in iterates keys, for-of iterates values.

jQuery

Key Points

- Button click: `$("#btn").click(()=>$("#p1").text("Hello"));`
- Factorial: calculate on change.
- Animation: `$("#div").animate({left:"200px"})`;
- Stop animation: `$("#div").stop()`;

Cross Q&A

- Q: How to select element in jQuery? A: Use `$()` with selector.
- Q: How to stop animation? A: Use `.stop()`.

AJAX

Key Points

- Load data: `$.get("url",(data)=>{})`.
- POST: `$.post("url",{title:"abc"},(res)=>{})`.

Cross Q&A

- Q: Difference between GET and POST? A: GET sends data in URL, POST sends in body.

- Q: How to load JSON data in table? A: Loop through data and append rows.

Extra Assignments

Key Points

- Password Generator: random string using `Math.random()`.
- Username Generator: first 2 chars of firstname + age + last 2 chars of lastname.

Cross Q&A

- Q: How to generate random password? A: Use `Math.random()` with character set.
- Q: How to calculate age from birthdate? A: Subtract birth year from current year, adjust for month/day.

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 - CSS: inline, internal, external.
 - JS: variables, functions, DOM, events, arrays, objects, JSON.
 - Node.js: callbacks, promises, `async/await`, `fs` module.
 - jQuery: selectors, events, animation.
 - AJAX: async data load, GET/POST.
 - Validation: HTML5 attributes, regex.
-

HTML & CSS

Key Points

- Inline CSS: style directly in tag.
- Internal CSS: inside `<style>` block.
- External CSS: separate `.css` file linked with `<link>`.
- Common properties: `font-family`, `font-size`, `font-style`, `font-weight`, `text-decoration`, `color`, `background-color`, `margin`, `border`.

- iframe: embed another page inside current page.
- Forms: `<input type="text" required minlength="8" maxlength="12">`.

Cross Q&A

- Q: Difference between inline and internal CSS? A: Inline is applied directly in the tag, internal is inside `<style>` in the HTML file.
- Q: How to open a link inside the same page? A: Use `<iframe>` and set hyperlink target to the iframe.
- Q: How to make an image circular? A: Use `border-radius:50%`.

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Cross Q&A

- Q: How to validate email in JS? A: Use regex pattern.
- Q: How to display array in UL? A: Loop through array and append `<li>` elements.
- Q: How to print multiplication table? A: Use a loop from 1 to 10 multiplying the number.

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- Q: How to convert currency? A: Multiply/divide based on conversion rate.

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- Q: How to add function to object prototype? A: `Object.prototype.print=function(){...}`.

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